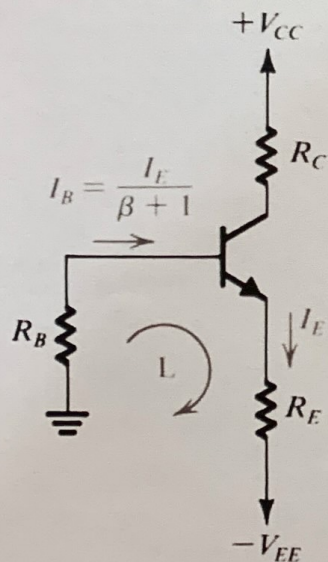


D7.36 The bias arrangement of Fig. 7.55 is to be used for a common-base amplifier. Design the circuit to establish a dc emitter current of 1 mA and provide the highest possible voltage gain while allowing for a signal swing at the collector of ± 2 V. Use +10-V and -5-V power supplies.

Ans. $R_B = 0$; $R_E = 4.3 \text{ k}\Omega$; $R_C = 8.4 \text{ k}\Omega$



R_B 析并到 E 极会降低增益. $\therefore$ 为了获得最大增益, $R_B = 0$

$$R_E = \frac{5 - 0.7}{1 \text{ mA}} = 4.3 \text{ k}\Omega$$

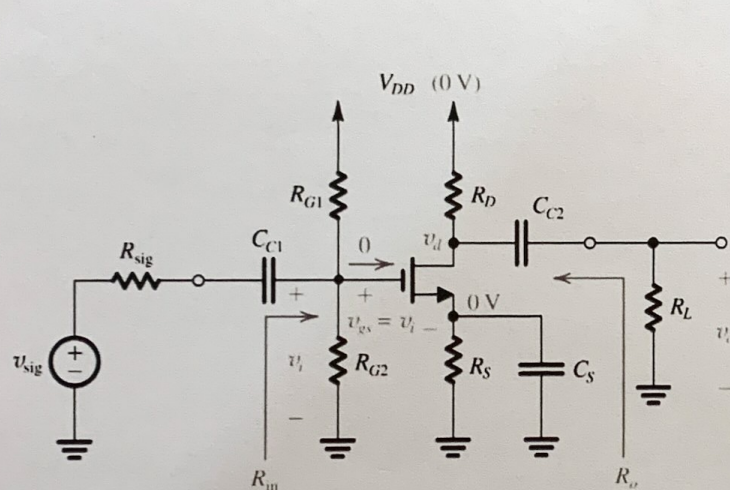
$A_v = g_m R_C \therefore R_C$ 要尽可能大, 也即 C 点电压要尽可能低

C 点电压下限为 $V_B - 0.4 = -0.4 \text{ V}$.
摆幅 $\pm 2 \text{ V}$ } $\Rightarrow V_C = -0.4 + 2 = 1.6 \text{ V}$

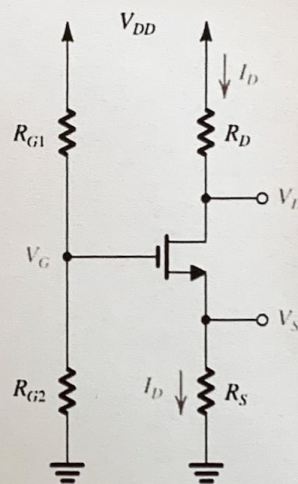
$$R_C = \frac{V_{CC} - V_C}{1 \text{ mA}} = 8.4 \text{ k}\Omega$$

D7.38 Design the bias circuit in Fig. 7.57(b) for the CS amplifier of Fig. 7.57(a). Assume the MOSFET is specified to have $V_t = 1$ V, $k_n = 4$ mA/V², and $V_A = 100$ V. Neglecting the Early effect, design for $I_D = 0.5$ mA, $V_S = 3.5$ V, and $V_D = 6$ V using a power-supply $V_{DD} = 15$ V. Specify the values of R_S and R_D . If a current of 2 μ A is used in the voltage divider, specify the values of R_{G1} and R_{G2} . Give the values of the MOSFET parameters g_m and r_o at the bias point.

Ans. $R_S = 7$ k Ω ; $R_D = 18$ k Ω ; $R_{G1} = 5$ M Ω ; $R_{G2} = 2.5$ M Ω ; $g_m = 2$ mA/V; $r_o = 200$ k Ω



(a)



(b)

$$R_S = \frac{3.5 \text{ V}}{0.5 \text{ mA}} = 7 \text{ k}\Omega$$

$$R_D = \frac{15 - 6}{0.5} = 18 \text{ k}\Omega$$

$$I_D = \frac{1}{2} k_n V_{ov}^2 \Rightarrow V_{ov} = \sqrt{\frac{2 I_D}{k_n}} = \frac{1}{2} \text{ V}$$

$$V_{GS} = V_{ov} + V_t = 1.5 \text{ V} \Rightarrow V_G = 5 \text{ V}$$

$$\therefore R_{G1} : R_{G2} = 2 : 1 \quad \left\{ \begin{array}{l} R_{G1} = 5 \text{ M}\Omega \\ R_{G2} = 2.5 \text{ M}\Omega \end{array} \right.$$

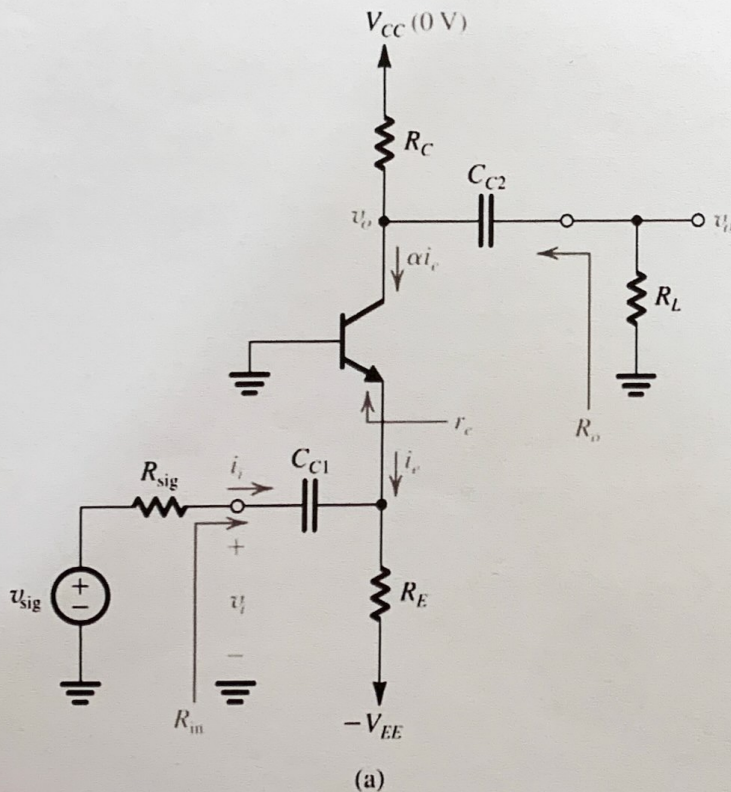
$$\frac{15}{R_{G1} + R_{G2}} = 2 \mu\text{A}$$

$$g_m = \frac{I_D}{V_{ov}/2} = \frac{0.5 \text{ mA}}{0.25 \text{ V}} = 2 \text{ mA/V}$$

$$r_o = \frac{V_A}{I_D} = \frac{100 \text{ V}}{0.5 \text{ mA}} = 200 \text{ k}\Omega$$

D7.44 Design the CB amplifier of Fig. 7.60(a) to provide an input resistance R_{in} that matches the source resistance of a cable with a characteristic resistance of $50\ \Omega$. Assume that $R_E \gg r_e$. The available power supplies are $\pm 5\text{ V}$ and $R_L = 8\text{ k}\Omega$. Design for a dc collector voltage $V_C = +1\text{ V}$. Specify the values of R_C and R_E . What is the overall voltage gain? If v_{sig} is a sine wave with a peak amplitude of 10 mV , what is the peak amplitude of the output voltage? Let $\alpha \simeq 1$.

Ans. $R_C = 8\text{ k}\Omega$; $R_E = 8.6\text{ k}\Omega$; 40 V/V ; 0.4 V



$$\frac{1}{g_m} = 50\ \Omega \Rightarrow g_m = \frac{1}{50}\ \text{A/V}$$

$$g_m = \frac{I_C}{V_T} \Rightarrow I_C = 0.5\text{ mA}$$

$$R_C = \frac{5 - 1}{0.5\text{ mA}} = 8\text{ k}\Omega$$

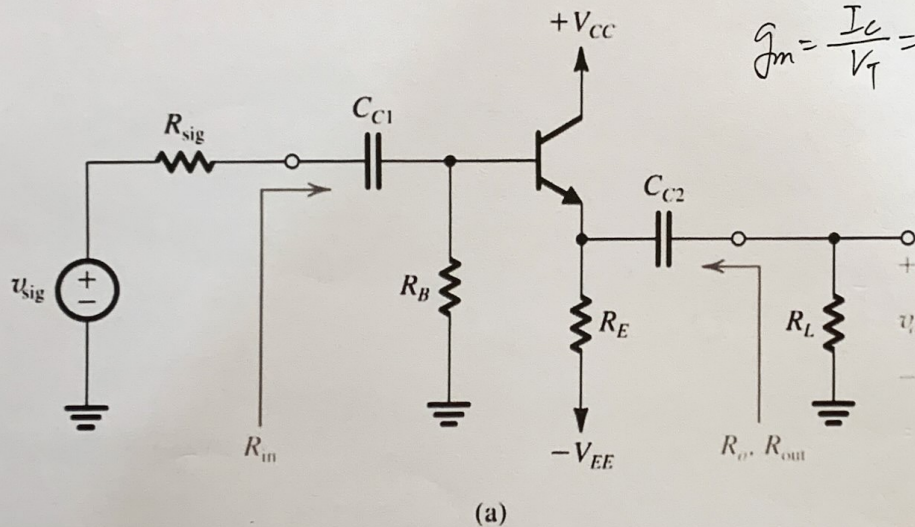
$$R_E = \frac{5 - 0.7}{0.5\text{ mA}} = 8.6\text{ k}\Omega$$

$$G_V = \frac{1}{2} \cdot g_m (R_C \parallel R_L) = \frac{1}{2} \times \frac{1}{50} \times 4\text{ k} = 40\text{ V/V}$$

$$\hat{v}_{out} = 10\text{ mV} \cdot G_V = 400\text{ mV}$$

D7.45 Design the emitter follower of Fig. 7.61(a) to operate at a dc emitter current $I_E = 1$ mA. Allow a dc voltage drop across R_B of 1 V. The available power supplies are ± 5 V, $\beta = 100$, $V_{BE} = 0.7$ V, and $V_A = 100$ V. Specify the values required for R_B and R_E . If $R_{sig} = 50$ k Ω and $R_L = 1$ k Ω , find R_{in} , v_i/v_{sig} , v_o/v_i , G_v , and R_{out} . (Note: In the bias design, neglect the Early effect.)

Ans. $R_B = 100$ k Ω ; $R_E = 3.3$ k Ω ; 44.3 k Ω ; 0.469 V/V; 0.968 V/V; 0.454 V/V; 320 Ω



$$g_m = \frac{I_C}{V_T} = 0.04 \text{ V/V} \quad R_E = \frac{5 - 1.7}{1 \text{ mA}} = 3.3 \text{ k}\Omega$$

$$R_B = \frac{1}{10 \mu\text{A}} = 100 \text{ k}\Omega$$

$$R_{in} = R_B \parallel (R_E \parallel R_L + \frac{1}{g_m}) \cdot (\beta + 1) \\ = 100 \text{ k} \parallel 80 \text{ k} = 44.4 \text{ k}$$

$$\frac{v_i}{v_{sig}} = \frac{R_{in}}{R_{in} + R_{sig}} = 0.47 \text{ V/V}$$

$$G_v = \frac{v_o}{v_{sig}} = \frac{R_E \parallel R_L}{R_E \parallel R_L + R_{in}(\beta + 1)} = \frac{767}{767 + 4400} = 0.147$$

$$\frac{v_o}{v_i} = \frac{R_E \parallel R_L}{R_E \parallel R_L + \frac{1}{g_m}} = \frac{767}{767 + 25} = 0.968 \text{ V/V}$$

$$G_v = \frac{v_o}{v_{sig}} = 0.455 \text{ V/V}$$

$$R_{out} = \left(\frac{R_{sig} \parallel R_B}{\beta + 1} + \frac{1}{g_m} \right) \parallel R_E \\ = 355 \parallel 3.3 \text{ k} \\ = 320 \Omega$$